

10.5.3 Gradients and Directional Derivatives

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$$f(x, y)$$

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \quad \text{Gradient is a vector}$$

Directional Derivative

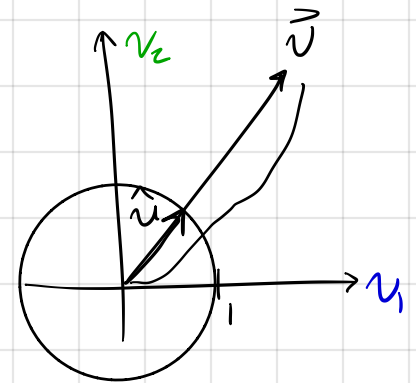
$$D_{\hat{u}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \hat{u}$$

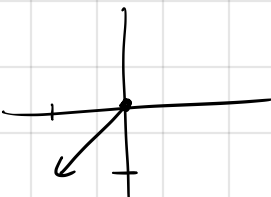
$$\|\hat{u}\| = 1$$

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\|\vec{v}\| = \sqrt{(v_1)^2 + (v_2)^2}$$

$$\hat{u} = \frac{\vec{v}}{\|\vec{v}\|}$$



27. $f(x, y) = e^{x+y}$ $(0, 0)$ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$ 

$\nabla f(x, y) = \begin{bmatrix} e^{x+y} \\ e^{x+y} \end{bmatrix}$ $\nabla f(0, 0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\|\vec{v}\| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$ $\hat{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

$\nabla f(0, 0) \cdot \hat{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = 1 \cdot \frac{-1}{\sqrt{2}} + 1 \cdot \frac{-1}{\sqrt{2}} = \boxed{\frac{-2}{\sqrt{2}}}$
 scalar

10.6. Maxima and Minima

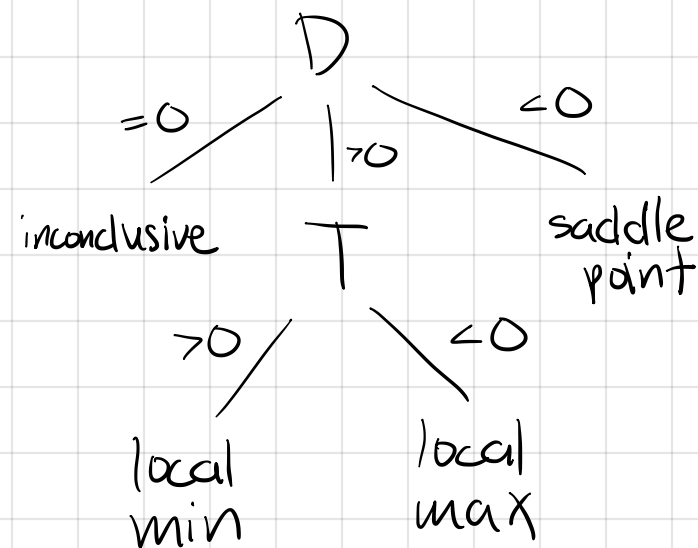
$$f(x, y) \quad (x_0, y_0)$$

$$\nabla f(x_0, y_0) = 0 = \vec{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = (0, 0)$$

$$\text{Hess } f(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

$$D = f_{xx}f_{yy} - (f_{xy})^2$$

$$T = f_{xx} + f_{yy}$$



10.6.

$$s. f(x, y) = -2x^2 + y^2 - 6y$$

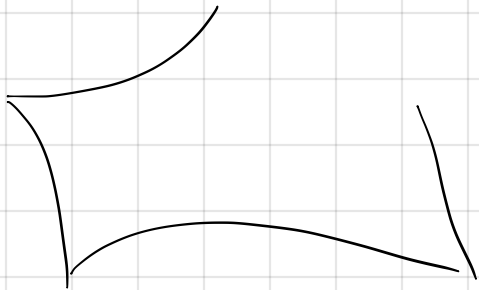
$$\nabla f(x, y) = \begin{bmatrix} -4x \\ 2y - 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{array}{l} \textcircled{1} -4x = 0 \Rightarrow x = 0 \\ \textcircled{2} 2y - 6 = 0 \Rightarrow y = 3 \end{array}$$

$(0, 3)$

$$\text{Hess } f(x, y) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\text{Hess } f(0, 3) = \begin{bmatrix} -4 & 0 \\ 0 & 2 \end{bmatrix} \quad D = (-4)(2) - (0)(0) = -8 < 0$$

saddle point @ $(0, 3)$



$$f(x, y) = xy - 2y^2$$

$$\nabla f(x, y) = \begin{bmatrix} y \\ x - 4y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\textcircled{1} y = 0$$

$$\textcircled{2} x - 4y = 0$$

$$(0, 0)$$

$$y = 0$$

$$x - 4(0) = 0$$

$$x = 0$$

$$\begin{bmatrix} 0 & 1 \\ 1 & -4 \end{bmatrix}$$

$$D = 0 \cdot (-4) - (1)(1) = -1 < 0 \Rightarrow \text{saddle point @ } (0, 0)$$

$$8. f(x,y) = yxe^{-y} = \overset{f_1(x,y)}{y} \cdot \overset{f_2(x,y)}{xe^{-y}} \quad (fg)' = f'g + g'f$$

$$\frac{\partial f}{\partial x} = ye^{-y} \quad \frac{\partial f}{\partial y} = (y)'xe^{-y} + (xe^{-y})'y = 1xe^{-y} - xe^{-y} \cdot y$$

$$= xe^{-y} - xye^{-y}$$

$$= (xy)'e^{-y} + (e^{-y})'xy$$

$$\begin{bmatrix} ye^{-y} \\ xe^{-y} - xye^{-y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{matrix} ye^{-y} = 0 & \textcircled{1} \\ xe^{-y} - xye^{-y} = 0 & \textcircled{2} \end{matrix}$$

$$\textcircled{1} ye^{-y} = 0$$

$$\swarrow \quad \searrow$$

$$y=0 \quad \cancel{e^{-y}=0}$$

$$(*) f(x)g(x) = 0$$

$$f(x)=0 \text{ or } g(x)=0$$

$$\text{Eg. } (x-1)(x-2)=0$$

$$\textcircled{2} xe^{-y} - xye^{-y}$$

$$xe^{-0} - x \cdot 0e^{-y} = 0$$

$$x \cdot 1 - 0 = 0$$

$$x = 0$$

$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = ye^{-y} \quad \frac{\partial f}{\partial y} = (y)'xe^{-y} + (xe^{-y})'y = 1 \cdot xe^{-y} - xe^{-y} \cdot y = xe^{-y} - xye^{-y}$$

$$f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (ye^{-y}) = 0$$

$$f_{yy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (xe^{-y} - xye^{-y}) = -xe^{-y} - (xe^{-y} - xye^{-y})$$

$$f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (xe^{-y} - xye^{-y}) = e^{-y} - ye^{-y} \quad f_{xy} = f_{yx}$$

$$f_{xx}(0,0) = 0$$

$$f_{yy}(0,0) = -0 \cdot e^{-0} - (0 \cdot e^{-0} - 0 \cdot 0e^{-0}) = 0$$

$$f_{xy}(0,0) = e^{-0} - 0 \cdot e^{-0} = 1$$

$$f_{xx}f_{yy} - (f_{xy})^2 = D$$

$$0 \cdot 0 - (1)^2 = -1 < 0$$

saddle point

$$a. f(x, y) = x \cos y$$

$$\nabla f(x, y) = \begin{bmatrix} \cos y \\ -x \sin y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{array}{l} \cos y = 0 \\ -x \sin y = 0 \end{array}$$

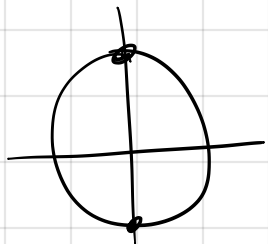
$$\textcircled{1} \cos y = 0$$

$$y = \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$$

$$\textcircled{2} -x \sin y = 0$$

$$-x(-1)^n = 0$$

$$x = 0$$



, $(0, -\frac{\pi}{2})$, $(0, \frac{\pi}{2})$, $(0, \frac{3\pi}{2})$, ... all saddle points

$$\begin{bmatrix} 0 & -\sin y \\ -\sin y & \underbrace{-x \cos y}_{=0} \end{bmatrix}$$

$$0 \cdot 0 - (-\sin y)^2$$

$$\underbrace{((-1)^n)^2}_{0 - 1 < 0}$$